# Software Testing

# Outline

- Overview of Software Testing
- Black Box / Functional Testing
- White Box / Structural Testing
- Integration Testing
- System Testing

# Overview of Software Testing

# Software Testing: Definition

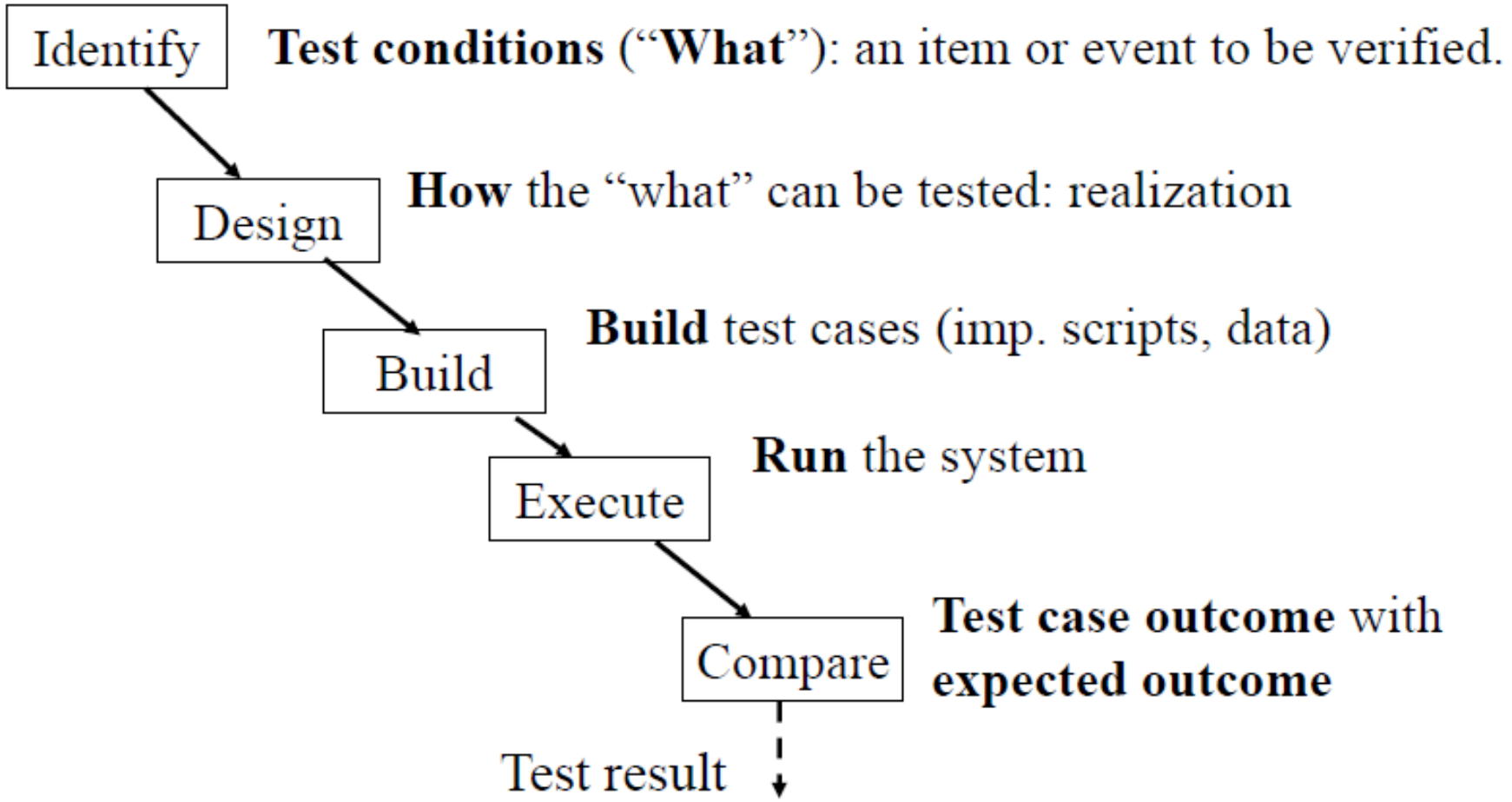
- Software testing is the process of evaluating and verifying that a software application or system meets its requirements and functions as expected https://www.ibm.com/think/topics/software-testing

- Testing can never establish the correctness of computer software. It can only find defects, not prove that there are none.”

# Why Testing?

- To discover more problems
  - Meyer's definition: "Software testing is a process of finding the errors in the product"
  - Testing never ends
- To ensure quality
  - Check whether the product matches with the requirements

# Testing Activities



# Test Stopping Criteria

- Meet deadline, exhaust budget, … (management decision)
- Achieved desired coverage
- Achieved desired level of failure intensity

# Black Box / Functional Testing

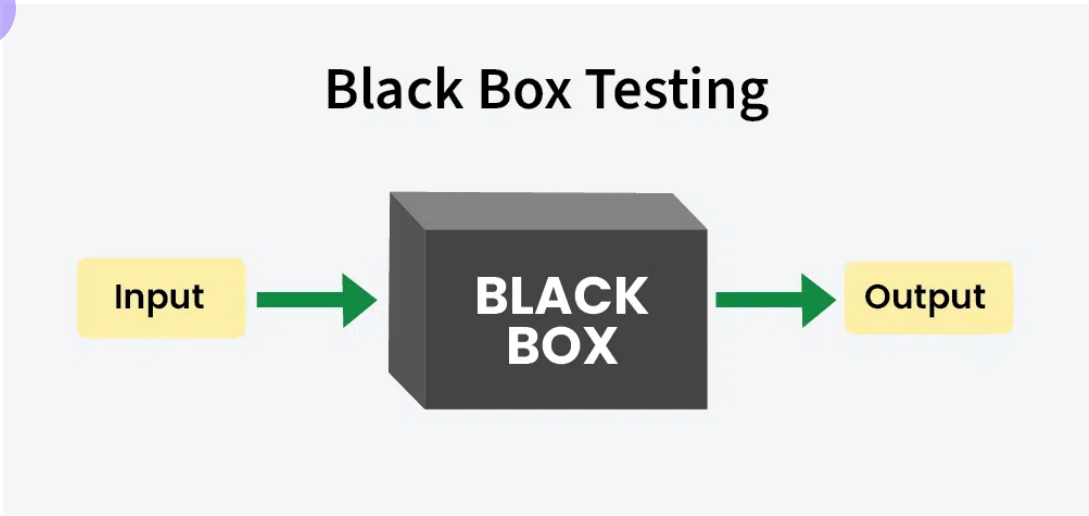
测试人员不看内部代码，通过 输入 → 观察结果验证

# Black-box Testing

- The only information used is the specification of the software.
- Functional test cases have two distinct advantages :
  - They are independent of how the software is developed, so if the implementation changes, the test cases are still useful;
  - Test case development can occur in parallel with the implementation, thereby reducing overall project development interval;
- Functional test cases suffer from two problems :
  - Significant redundancies may exist among test cases;
  - Compounded by the possibility of gaps of untested software;

# Definition of Black-box Testing

- **Black-box testing** is a software testing methodology where the system under test is evaluated based on its external behavior, without knowledge of its internal code, structure, or implementation details. Testers focus on inputs and outputs, validating whether the software functions according to specified requirements, user expectations, and desired outcomes.



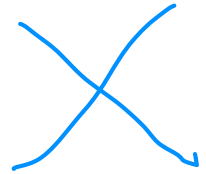
# Black-box Testing

- Focus: I/O behavior. If for any given input, we can predict the output, then the module passes the test.
  - Almost always impossible to generate all possible inputs ("test cases")
- Goal: Reduce number of test cases by equivalence partitioning:
  - Divide input conditions into equivalence classes
  - Choose test cases for each equivalence class. (Example: If an object is supposed to accept a negative number,  testing one negative number is enough)

# Black-box Testing 常用方法

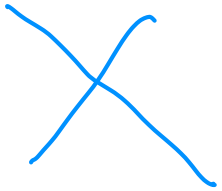
- **Boundary Value Testing (BVT)**
  - Normal boundary value testing
  - Robust boundary value testing
  - Worst-case boundary value testing
  - Robust worst-case boundary value testing
- **Equivalence Class Testing**
  - Weak Normal Equivalence Class Testing
  - Strong Normal Equivalence Class Testing
  - Weak Robust Equivalence Class Testing
  - Strong Robust Equivalence Class Testing
- **Decision Table Based Testing**

# Boundary Value Testing : Overview



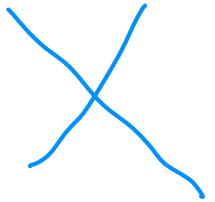
- Any program can be considered to be a function
  - Program inputs form its domain
  - Program outputs form its range
- Boundary value analysis is the best known functional testing technique.
- The objective of functional testing is to use knowledge of the functional nature of a program to identify test cases.
- Historically, functional testing has focused on the input domain, but it is a good supplement to consider test cases based on the range (outputs) as well.

# Boundary Value Testing: Overview



- Boundary value testing focuses on the boundary of the input space to identify test cases.
- The rationale behind boundary value analysis is that errors tend to occur near the extreme values of an input variable.
- Programs written in not strongly typed languages are more appropriate candidates for boundary value testing

# Boundary Value Testing: Overview



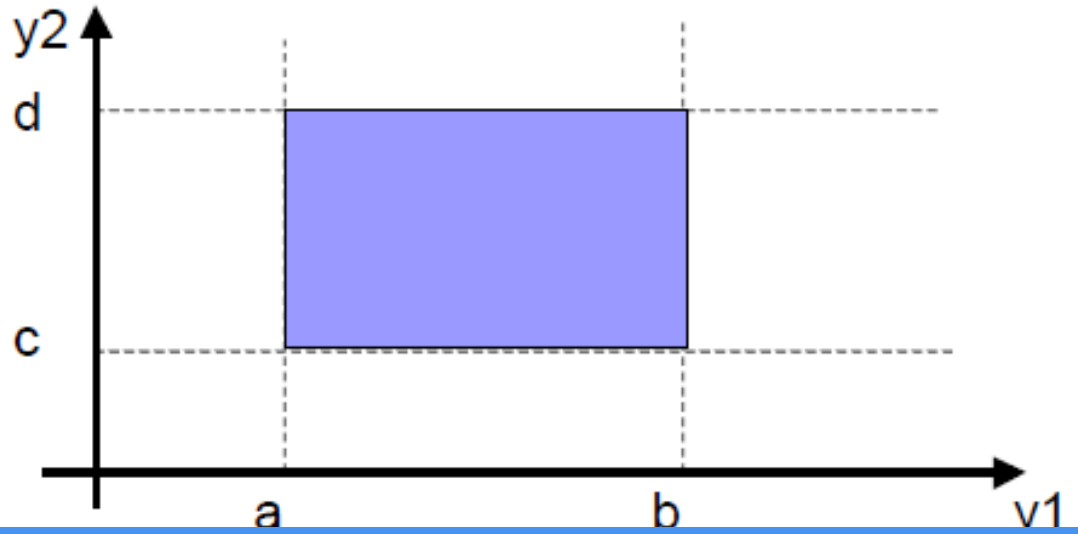
- BVT tends to generate more test cases with poorer test coverage than domain or equivalence testing.
- Due to its simplicity, BVT test case generation can be easily automated.
- In our discussion we will assume a program P accepting two inputs y1 and y2 such that $a \leq y1 \leq b$ and $c \leq y2 \leq d$

# Valid Input for Program P

- consider the following function:

$$f(y_1, y_2), \text{ where } a \le y_1 \le b\ , c \le y_2 \le d$$

- boundary inequalities of n input variables define an n-dimensional input space (domain):



# Variant of Boundary Value Testing

- Two independent considerations: -
  - Concerned with invalid values of variables?
    - Only valid values – Normal
    - Both valid and invalid values – Robust
  - Single fault assumption?
    - Fault are due to invalidity of single variable.
    - Fault are due to the interaction among two or more variables.
- Two considerations yield 4 type of boundary value testing: -
  - Normal boundary value testing
  - Robust boundary value testing
  - Worst-case boundary value testing
  - Robust worse-case value testing

# ① Normal Boundary Value Testing 正常边界值测试

- The basic idea in boundary value analysis is to select input variable values at their:
  - Minimum
  - Just above the minimum
  - A nominal value ( (minmum+maximum)/2)
  - Just below the maximum
  - Maximum
- We use the conventions (min, min+, nom, max-, and max) to refer to these values

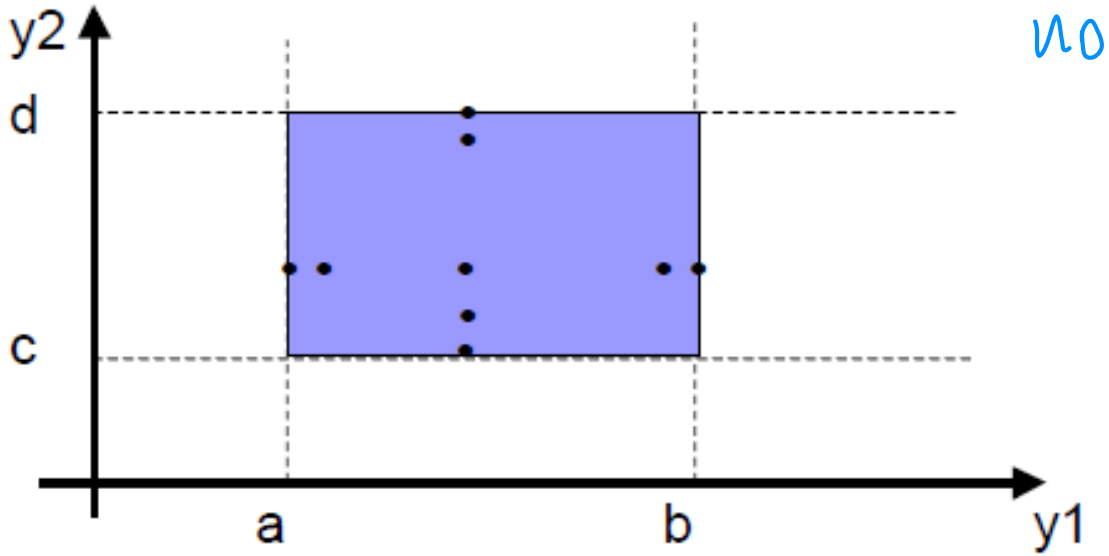
# Normal Boundary Value Testing

- Augmented by the **single fault assumption** principle.

> "Failures occur rarely as the result of the simultaneous occurrence of two (or more) faults"

- In this respect, normal boundary value test cases can be obtained by holding all but one at the nominal values and let the remaining variable assume the min, min+, nom, max–, and max values, repeating this for each variable. Thus, for a function of n variables, boundary value analysis yields 4n + 1 unique test cases.

# Normal Boundary Value Testing for Program P




$$T = \{ \langle y_{1nom}, y_{2min} \rangle, \langle y_{1nom}, y_{2min+} \rangle, \langle y_{1nom}, y_{2nom} \rangle,$$
$$\langle y_{1nom}, y_{2max-} \rangle, \langle y_{1nom}, y_{2max+} \rangle, \langle y_{1min}, y_{2nom} \rangle,$$
$$\langle {}_{1nin+}, y_{2nom} \rangle, \langle y_{1max-}, y_{2nom} \rangle, \langle y_{1max}, y_{2nom} \rangle \}$$

# Generalizing Boundary Value Analysis

- Boundary value analysis is a test design technique used to identify and select input or output values.
- The basic boundary value analysis for normal boundary value testing can be generalized in two ways:
  - By the number of variables - (4n +1) test cases for n variables
  - By the kinds of ranges of variables
    - Bounded discrete variable – 5 test values (min, min+, nom, max-, max)
    - Unbounded discrete (no upper or lower bounds clearly defined), create “artificial” bounds
    - Logical variables present a problem for boundary value analysis, only true and false

# Limitations of Boundary Value Analysis

- Boundary value analysis works well when the program to be tested is a function of several independent variables that represent bounded physical quantities.
- Boundary value analysis selected test data with no consideration of the function of the program, nor of the semantic meaning of the variables.
- We shall distinguish between physical and logical type of variables (e.g. temperature, pressure speed, or PIN numbers, telephone numbers etc.) . BVT works well for physical variables but not for logical variables.

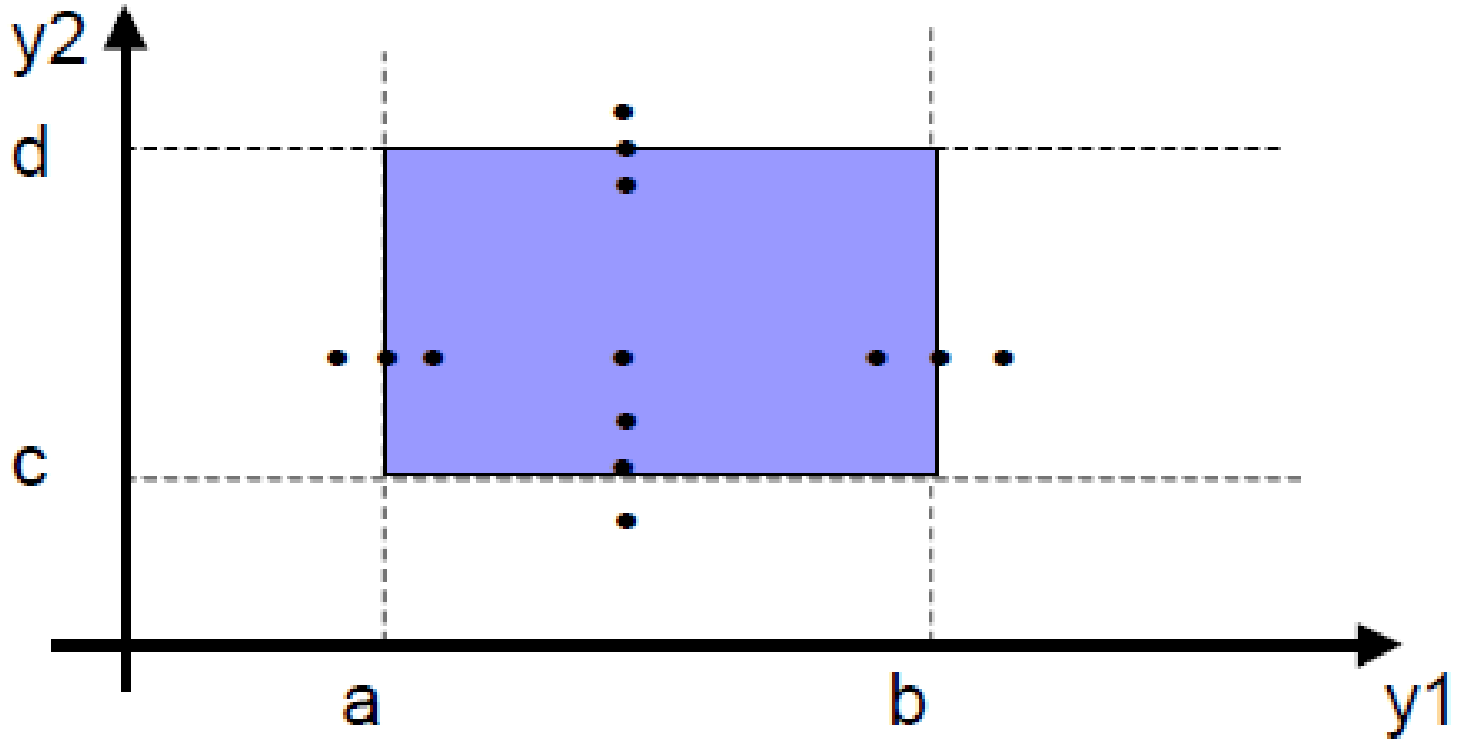
# 2 Robust Boundary Value Testing

- Robustness testing is a simple extension of boundary value analysis.
- In addition to the five boundary value analysis values of variables, we add values slightly greater that the maximum (max+) and a value slightly less than the minimum (min-).
- In this respect, for a function of n variables, there will be 6n + 1 unique test cases.
- The main value of robustness testing is to force attention on exception handling.

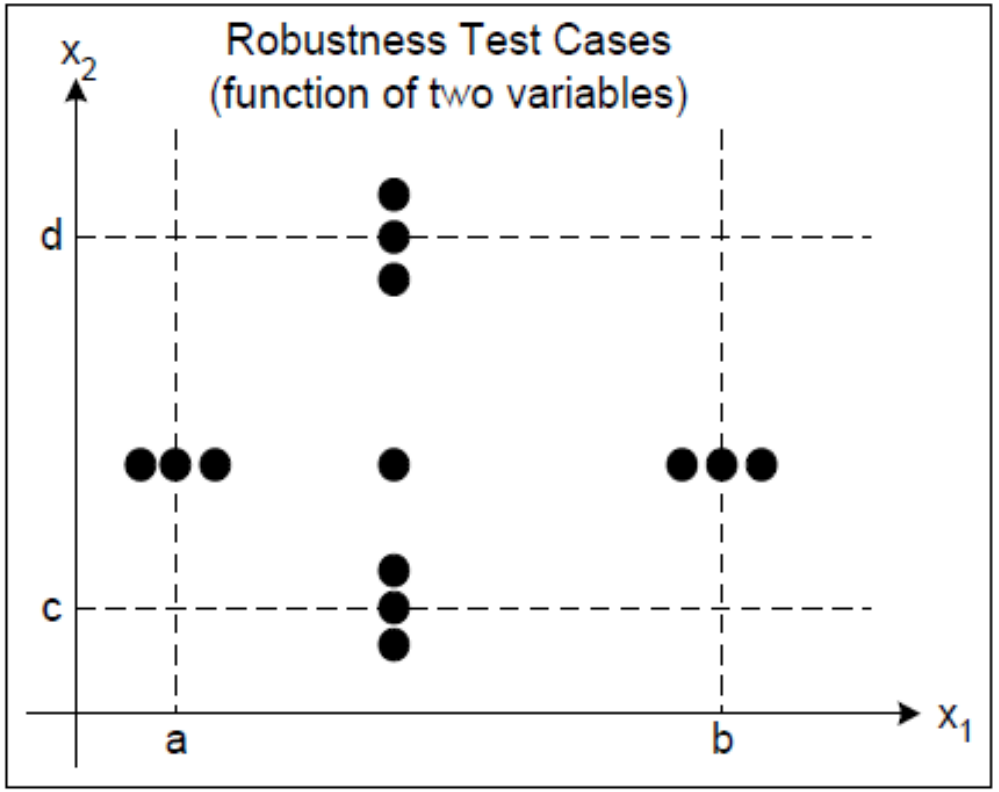
# Robust Boundary Value Testing

- In some strongly typed languages values beyond the predefined range will cause a run-time error.
- It is a choice of using a weak typed language with exception handling or a strongly typed language with explicit logic to handle out of range values.

# Robustness Test Cases for Program P



# Robustness Test Cases for Program P




$\langle x_{1nom}, x_{2min} \rangle$
$\langle x_{1nom}, x_{2min+} \rangle$
$\langle x_{1nom}, x_{2nom} \rangle$
$\langle x_{1nom}, x_{2max-} \rangle$
$\langle x_{1nom}, x_{2max} \rangle$
$\langle x_{1min}, x_{2nom} \rangle$
$\langle x_{1min+}, x_{2nom} \rangle$
$\langle x_{1nom}, x_{2nom} \rangle$
$\langle x_{1max-}, x_{2nom} \rangle$
$\langle x_{1max}, x_{2nom} \rangle$
$\langle x_{1min-}, x_{2nom} \rangle$
$\langle x_{1nom}, x_{2max+} \rangle$
$\langle x_{1max+}, x_{2nom} \rangle$
$\langle x_{1nom}, x_{2min-} \rangle$

# 3 Worst-case Boundary Value Testing

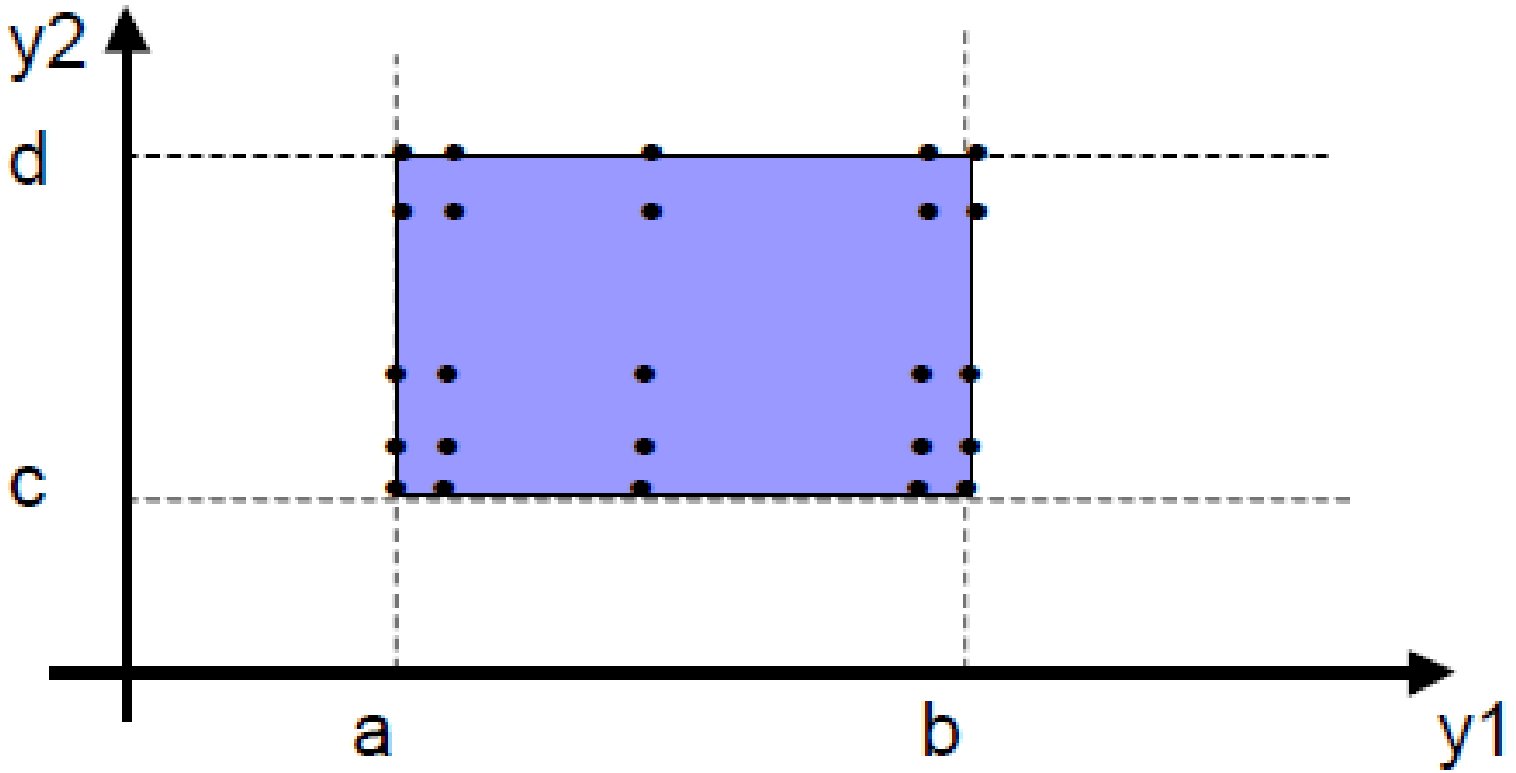
- In worst case testing we reject the single fault assumption and we are interested in what happens when more than one variable has an extreme value.

- Considering that we have five different values that can be considered during normal boundary value testing for one variable, now we take the Cartesian product of these possible values for 2, 3, … n variables. In this respect we can have $5^n$ test cases for n input variables.

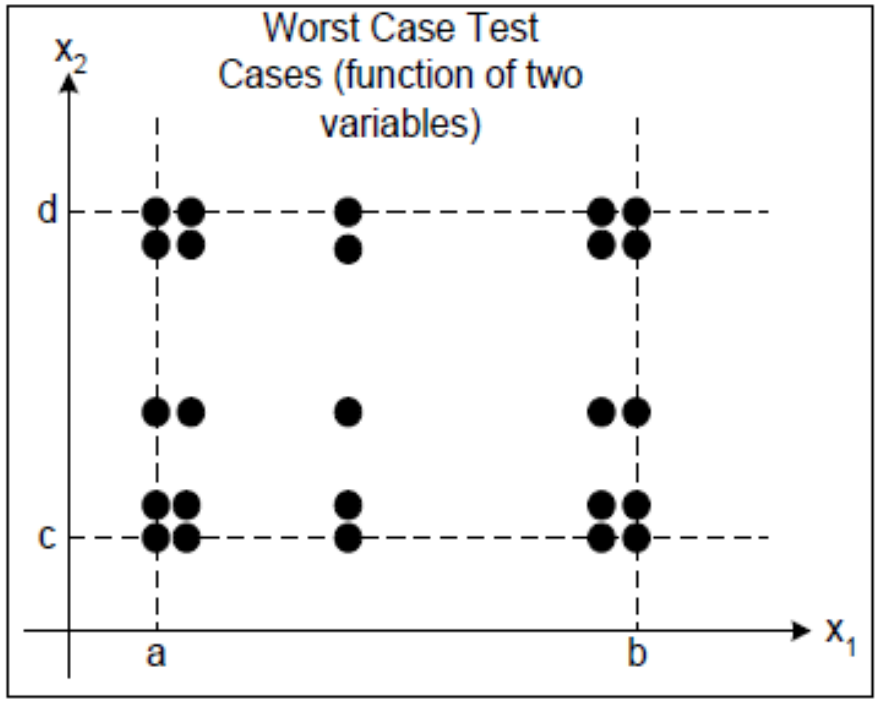
一个变量5个值

- The best application of worst case testing is where physical variables have numerous interactions and failure of a program is costly.

# Worst-case Boundary Value Testing for Program P



# Worst-case Boundary Value Testing for Program P




$\langle x_{1min}, x_{2min} \rangle$
$\langle x_{1min}, x_{2min+} \rangle$
$\langle x_{1min}, x_{2nom} \rangle$
$\langle x_{1min}, x_{2max-} \rangle$
$\langle x_{1min}, x_{2max} \rangle$
$\langle x_{1min+}, x_{2min} \rangle$
$\langle x_{1min+}, x_{2min+} \rangle$
$\langle x_{1min+}, x_{2nom} \rangle$
$\langle x_{1min+}, x_{2max-} \rangle$
$\langle x_{1min+}, x_{2max} \rangle$
$\langle x_{1nom}, x_{2min} \rangle$
$\langle x_{1nom}, x_{2min+} \rangle$
$\langle x_{1nom}, x_{2nom} \rangle$
$\langle x_{1nom}, x_{2max-} \rangle$

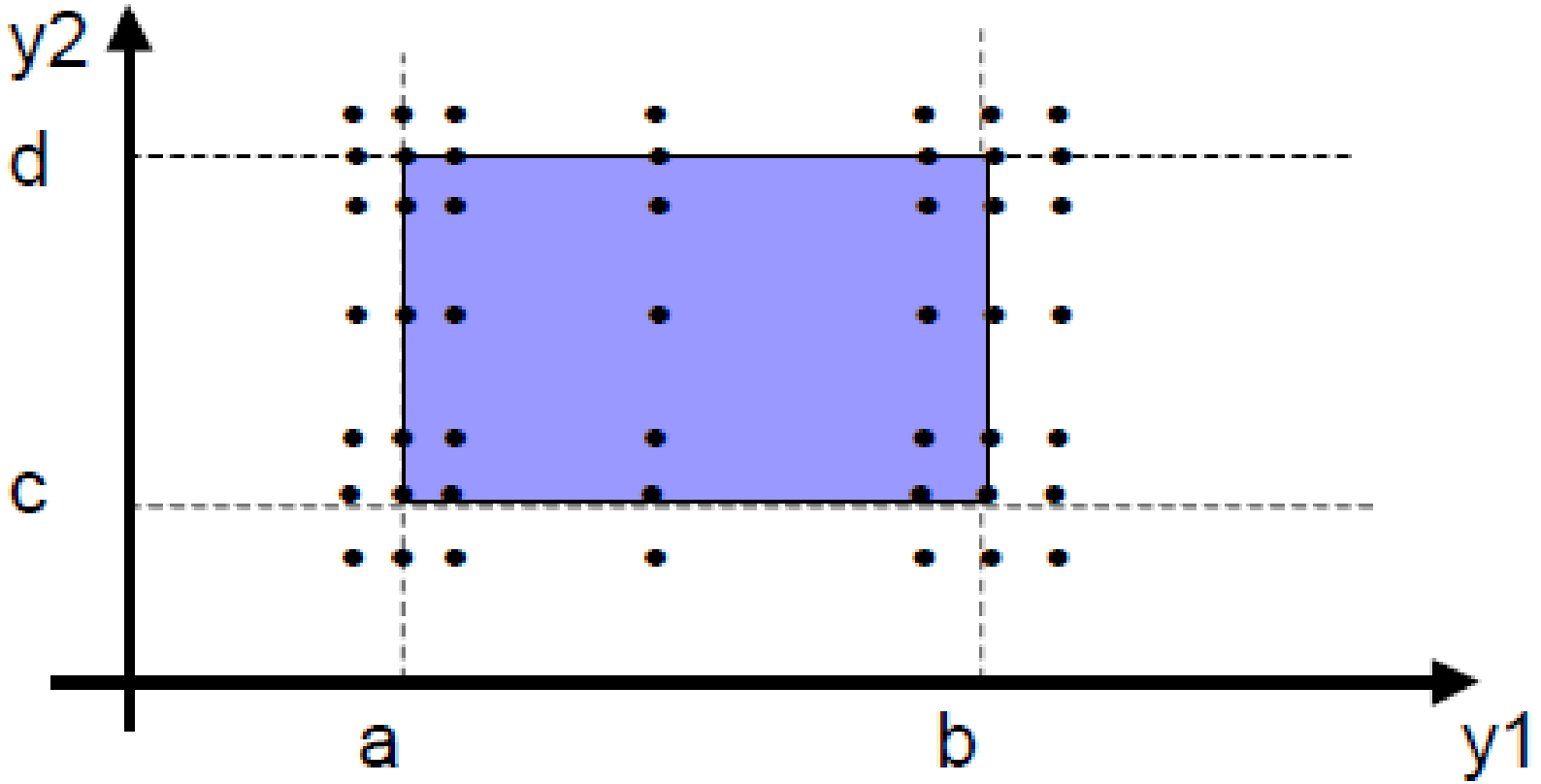
$\langle x_{1max-}, x_{2min} \rangle$
$\langle x_{1max-}, x_{2min+} \rangle$
$\langle x_{1max-}, x_{2nom} \rangle$
$\langle x_{1max-}, x_{2max-} \rangle$
$\langle x_{1max-}, x_{2max} \rangle$
$\langle x_{1max}, x_{2min} \rangle$
$\langle x_{1max}, x_{2min+} \rangle$
$\langle x_{1max}, x_{2nom} \rangle$
$\langle x_{1max}, x_{2max-} \rangle$
$\langle x_{1max}, x_{2max} \rangle$

# 4 Robust Worst-case Boundary Value Testing

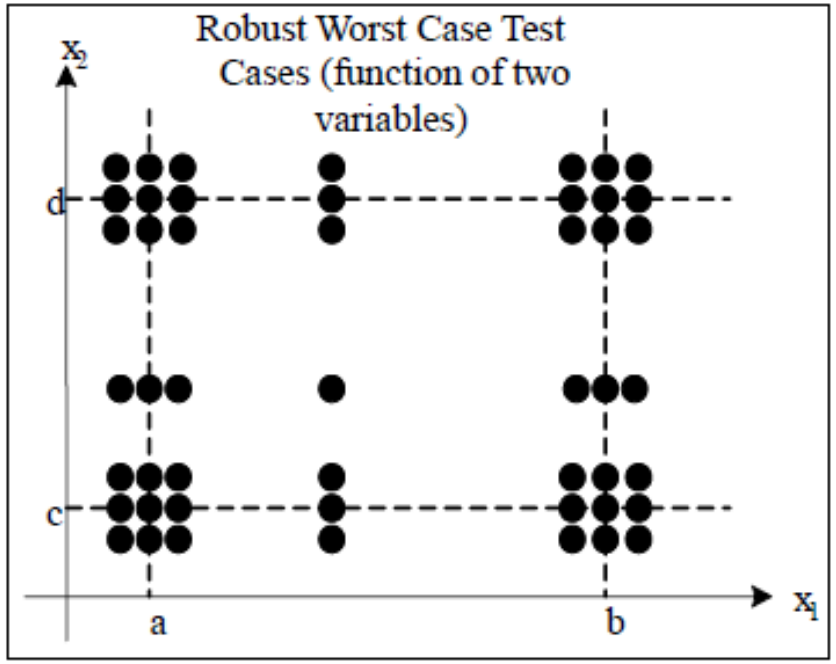
- Worst case testing can be further augmented by considering robust worst case testing (i.e. adding slightly out of bounds values to the five already considered). This involves the Cartesian product of the seven-element sets we used in robustness testing.

- In this respect we can have $7^n$ test cases for n input variables.

# Robust Worst-case Boundary Value Testing for Program P



# Robust Worst-case Boundary Value Testing for Program P

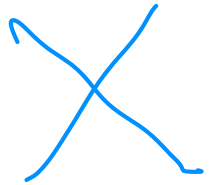



| | | | |
|---|---|---|---|
| $\langle X_{1min-}, X_{2min-} \rangle$ | $\langle X_{1min+}, X_{2min-} \rangle$ | $\langle X_{1max-}, X_{2min-} \rangle$ | $\langle X_{1max+}, X_{2min-} \rangle$ |
| $\langle X_{1min-}, X_{2min} \rangle$ | $\langle X_{1min+}, X_{2min} \rangle$ | $\langle X_{1max-}, X_{2min} \rangle$ | $\langle X_{1max+}, X_{2min} \rangle$ |
| $\langle X_{1min-}, X_{2min+} \rangle$ | $\langle X_{1min+}, X_{2min+} \rangle$ | $\langle X_{1max-}, X_{2min+} \rangle$ | $\langle X_{1max+}, X_{2min+} \rangle$ |
| $\langle X_{1min-}, X_{2nom} \rangle$ | $\langle X_{1min+}, X_{2nom} \rangle$ | $\langle X_{1max-}, X_{2nom} \rangle$ | $\langle X_{1max+}, X_{2nom} \rangle$ |
| $\langle X_{1min-}, X_{2max-} \rangle$ | $\langle X_{1min+}, X_{2max-} \rangle$ | $\langle X_{1max-}, X_{2max-} \rangle$ | $\langle X_{1max+}, X_{2max-} \rangle$ |
| $\langle X_{1min-}, X_{2max} \rangle$ | $\langle X_{1min+}, X_{2max} \rangle$ | $\langle X_{1max-}, X_{2max} \rangle$ | $\langle X_{1max+}, X_{2max} \rangle$ |
| $\langle X_{1min-}, X_{2max+} \rangle$ | $\langle X_{1min+}, X_{2max+} \rangle$ | $\langle X_{1max-}, X_{2max+} \rangle$ | $\langle X_{1max+}, X_{2max+} \rangle$ |
| $\langle X_{1min}, X_{2min-} \rangle$ | $\langle X_{1nom}, X_{2min-} \rangle$ | $\langle X_{1max}, X_{2min-} \rangle$ | |
| $\langle X_{1min}, X_{2min} \rangle$ | $\langle X_{1nom}, X_{2min} \rangle$ | $\langle X_{1max}, X_{2min} \rangle$ | |
| $\langle X_{1min}, X_{2min+} \rangle$ | $\langle X_{1nom}, X_{2min+} \rangle$ | $\langle X_{1max}, X_{2min+} \rangle$ | |
| $\langle X_{1min}, X_{2nom} \rangle$ | $\langle X_{1nom}, X_{2nom} \rangle$ | $\langle X_{1max}, X_{2nom} \rangle$ | |
| $\langle X_{1min}, X_{2max-} \rangle$ | $\langle X_{1nom}, X_{2max-} \rangle$ | $\langle X_{1max}, X_{2max-} \rangle$ | |
| $\langle X_{1min}, X_{2max} \rangle$ | $\langle X_{1nom}, X_{2max} \rangle$ | $\langle X_{1max}, X_{2max} \rangle$ | |

# Example – The Triangle Problem

- Problem Statement
  - Input: 3 integers (sides of a triangle)
  - Output: Type of Triangle (Equilateral, Isosceles, Scalene or NotATriangle)
  - Extended Version: Additional Output Type: Right Triangle

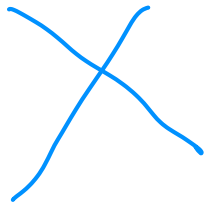
# Example – The Triangle Problem



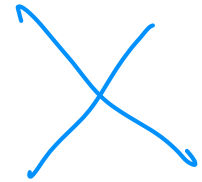
- In the problem statement, no conditions are specified on the triangle sides, other than being integers.
- Obviously, the lower bounds of the ranges are all 1.
- We arbitrarily take 200 as an upper bound.
- For each side, the test values are {1, 2, 100, 199, 200}.
- Robust boundary value test cases will add {0, 201}.
- The table on the next page contains Normal Boundary value test cases using these ranges.
  - Test cases 3, 8, and 13 are identical – Redundant
  - There is no test case for scalene triangles

# Example

| *Case* | a | b | c | *Expected Output* |
|---|---|---|---|---|
| 1 | 100 | 100 | 1 | Isosceles |
| 2 | 100 | 100 | 2 | Isosceles |
| 3 | 100 | 100 | 100 | Equilateral |
| 4 | 100 | 100 | 199 | Isosceles |
| 5 | 100 | 100 | 200 | Not a triangle |
| 6 | 100 | 1 | 100 | Isosceles |
| 7 | 100 | 2 | 100 | Isosceles |
| 8 | 100 | 100 | 100 | Equilateral |
| 9 | 100 | 199 | 100 | Isosceles |
| 10 | 100 | 200 | 100 | Not a triangle |
| 11 | 1 | 100 | 100 | Isosceles |
| 12 | 2 | 100 | 100 | Isosceles |
| 13 | 100 | 100 | 100 | Equilateral |
| 14 | 199 | 100 | 100 | Isosceles |
| 15 | 200 | 100 | 100 | Not a triangle |



# Example – The NextDate Function



- The function takes 3 variables, month, the day, and the year. It return the next day of the input.
- We could encode these, so that January would correspond to 1, February to 2, and so on.
- In this example, we use Worst-case boundary value testing. All 125 worst-case test cases for NextDate are listed in the table on next page.
- Examine the following: -
  - Gaps of untested functionality
  - Redundant testing
- Questions: -
  - Would anyone actually want to test January 1 in five different years?
  - Is the end of February tested sufficiently?

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1812 | 1, 2, 1812 |
| 2 | 1 | 1 | 1813 | 1, 2, 1813 |
| 3 | 1 | 1 | 1912 | 1, 2, 1912 |
| 4 | 1 | 1 | 2011 | 1, 2, 2011 |
| 5 | 1 | 1 | 2012 | 1, 2, 2012 |
| 6 | 1 | 2 | 1812 | 1, 3, 1812 |
| 7 | 1 | 2 | 1813 | 1, 3, 1813 |
| 8 | 1 | 2 | 1912 | 1, 3, 1912 |
| 9 | 1 | 2 | 2011 | 1, 3, 2011 |
| 10 | 1 | 2 | 2012 | 1, 3, 2012 |
| 11 | 1 | 15 | 1812 | 1, 16, 1812 |
| 12 | 1 | 15 | 1813 | 1, 16, 1813 |
| 13 | 1 | 15 | 1912 | 1, 16, 1912 |
| 14 | 1 | 15 | 2011 | 1, 16, 2011 |
| 15 | 1 | 15 | 2012 | 1, 16, 2012 |
| 16 | 1 | 30 | 1812 | 1, 31, 1812 |
| 17 | 1 | 30 | 1813 | 1, 31, 1813 |
| 18 | 1 | 30 | 1912 | 1, 31, 1912 |
| 19 | 1 | 30 | 2011 | 1, 31, 2011 |
| 20 | 1 | 30 | 2012 | 1, 31, 2012 |
| 21 | 1 | 31 | 1812 | 2, 1, 1812 |
| 22 | 1 | 31 | 1813 | 2, 1, 1813 |
| 23 | 1 | 31 | 1912 | 2, 1, 1912 |
| 24 | 1 | 31 | 2011 | 2, 1, 2011 |
| 25 | 1 | 31 | 2012 | 2, 1, 2012 |
| 26 | 2 | 1 | 1812 | 2, 2, 1812 |
| 27 | 2 | 1 | 1813 | 2, 2, 1813 |
| 30 | 2 | 1 | 2012 | 2, 2, 2012 |
| 31 | 2 | 2 | 1812 | 2, 3, 1812 |
| 32 | 2 | 2 | 1813 | 2, 3, 1813 |
| 33 | 2 | 2 | 1912 | 2, 3, 1912 |
| 34 | 2 | 2 | 2011 | 2, 3, 2011 |
| 35 | 2 | 2 | 2012 | 2, 3, 2012 |
| 36 | 2 | 15 | 1812 | 2, 16, 1812 |
| 37 | 2 | 15 | 1813 | 2, 16, 1813 |
| 38 | 2 | 15 | 1912 | 2, 16, 1912 |
| 39 | 2 | 15 | 2011 | 2, 16, 2011 |
| 40 | 2 | 15 | 2012 | 2, 16, 2012 |
| 41 | 2 | 30 | 1812 | Invalid date |
| 42 | 2 | 30 | 1813 | Invalid date |
| 43 | 2 | 30 | 1912 | Invalid date |
| 44 | 2 | 30 | 2011 | Invalid date |
| 45 | 2 | 30 | 2012 | Invalid date |
| 46 | 2 | 31 | 1812 | Invalid date |
| 47 | 2 | 31 | 1813 | Invalid date |
| 48 | 2 | 31 | 1912 | Invalid date |
| 49 | 2 | 31 | 2011 | Invalid date |
| 50 | 2 | 31 | 2012 | Invalid date |
| 51 | 6 | 1 | 1812 | 6, 2, 1812 |
| 52 | 6 | 1 | 1813 | 6, 2, 1813 |
| 53 | 6 | 1 | 1912 | 6, 2, 1912 |
| 54 | 6 | 1 | 2011 | 6, 2, 2011 |
| 55 | 6 | 1 | 2012 | 6, 2, 2012 |
| 56 | 6 | 2 | 1812 | 6, 3, 1812 |

| | | | | |
|---|---|---|---|---|
| 59 | 6 | 2 | 2011 | 6, 3, 2011 |
| 60 | 6 | 2 | 2012 | 6, 3, 2012 |
| 61 | 6 | 15 | 1812 | 6, 16, 1812 |
| 62 | 6 | 15 | 1813 | 6, 16, 1813 |
| 63 | 6 | 15 | 1912 | 6, 16, 1912 |
| 64 | 6 | 15 | 2011 | 6, 16, 2011 |
| 65 | 6 | 15 | 2012 | 6, 16, 2012 |
| 66 | 6 | 30 | 1812 | 7, 1, 1812 |
| 67 | 6 | 30 | 1813 | 7, 1, 1813 |
| 68 | 6 | 30 | 1912 | 7, 1, 1912 |
| 69 | 6 | 30 | 2011 | 7, 1, 2011 |
| 70 | 6 | 30 | 2012 | 7, 1, 2012 |
| 71 | 6 | 31 | 1812 | Invalid date |
| 72 | 6 | 31 | 1813 | Invalid date |
| 73 | 6 | 31 | 1912 | Invalid date |
| 74 | 6 | 31 | 2011 | Invalid date |
| 75 | 6 | 31 | 2012 | Invalid date |
| 76 | 11 | 1 | 1812 | 11, 2, 1812 |
| 77 | 11 | 1 | 1813 | 11, 2, 1813 |
| 78 | 11 | 1 | 1912 | 11, 2, 1912 |
| 79 | 11 | 1 | 2011 | 11, 2, 2011 |
| 80 | 11 | 1 | 2012 | 11, 2, 2012 |
| 81 | 11 | 2 | 1812 | 11, 3, 1812 |
| 82 | 11 | 2 | 1813 | 11, 3, 1813 |
| 83 | 11 | 2 | 1912 | 11, 3, 1912 |
| 84 | 11 | 2 | 2011 | 11, 3, 2011 |

| | | | | |
|---|---|---|---|---|
| 88 | 11 | 15 | 1912 | 11, 16, 1912 |
| 89 | 11 | 15 | 2011 | 11, 16, 2011 |
| 90 | 11 | 15 | 2012 | 11, 16, 2012 |
| 91 | 11 | 30 | 1812 | 12, 1, 1812 |
| 92 | 11 | 30 | 1813 | 12, 1, 1813 |
| 93 | 11 | 30 | 1912 | 12, 1, 1912 |
| 94 | 11 | 30 | 2011 | 12, 1, 2011 |
| 95 | 11 | 30 | 2012 | 12, 1, 2012 |
| 96 | 11 | 31 | 1812 | Invalid date |
| 97 | 11 | 31 | 1813 | Invalid date |
| 98 | 11 | 31 | 1912 | Invalid date |
| 99 | 11 | 31 | 2011 | Invalid date |
| 100 | 11 | 31 | 2012 | Invalid date |
| 101 | 12 | 1 | 1812 | 12, 2, 1812 |
| 102 | 12 | 1 | 1813 | 12, 2, 1813 |
| 103 | 12 | 1 | 1912 | 12, 2, 1912 |
| 104 | 12 | 1 | 2011 | 12, 2, 2011 |
| 105 | 12 | 1 | 2012 | 12, 2, 2012 |
| 106 | 12 | 2 | 1812 | 12, 3, 1812 |
| 107 | 12 | 2 | 1813 | 12, 3, 1813 |
| 108 | 12 | 2 | 1912 | 12, 3, 1912 |
| 109 | 12 | 2 | 2011 | 12, 3, 2011 |
| 110 | 12 | 2 | 2012 | 12, 3, 2012 |
| 111 | 12 | 15 | 1812 | 12, 16, 1812 |
| 112 | 12 | 15 | 1813 | 12, 16, 1813 |
| 113 | 12 | 15 | 1912 | 12, 16, 1912 |

| *Case* | *Month* | *Day* | *Year* | *Expected Output* |
|---|---|---|---|---|
| 116 | 12 | 30 | 1812 | 12, 31, 1812 |
| 117 | 12 | 30 | 1813 | 12, 31, 1813 |
| 118 | 12 | 30 | 1912 | 12, 31, 1912 |
| 119 | 12 | 30 | 2011 | 12, 31, 2011 |
| 120 | 12 | 30 | 2012 | 12, 31, 2012 |
| 121 | 12 | 31 | 1812 | 1, 1, 1813 |
| 122 | 12 | 31 | 1813 | 1, 1, 1814 |
| 123 | 12 | 31 | 1912 | 1, 1, 1913 |
| 124 | 12 | 31 | 2011 | 1, 1, 2012 |
| 125 | 12 | 31 | 2012 | 1, 1, 2013 |

# Black-box Testing

- Boundary Value Testing (BVT)
  - Normal boundary value testing
  - Robust boundary value testing
  - Worst-case boundary value testing
  - Robust worst-case boundary value testing
- **Equivalence Class Testing**
  - Weak Normal Equivalence Class Testing
  - Strong Normal Equivalence Class Testing
  - Weak Robust Equivalence Class Testing
  - Strong Robust Equivalence Class Testing
  - Decision Table Based Testing

# Equivalence Class Testing - Overview

- BVT assumes that input variables are independent of one another
- BVT tends to generate more test cases with poorer test coverage (with the independence assumption) than domain or equivalence testing.
- Equivalence classes form a partition of a set that is a collection of mutually disjoint subsets whose union is the entire set.
- Two important implications for testing:
  - The fact that the entire set is represented provides a form of completeness
  - The disjointness assures a form of non-redundancy

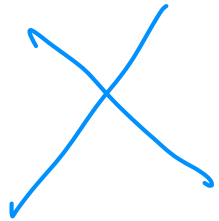
# Equivalence Classes

- The idea of equivalence class testing is to identify test cases by using one element from each equivalence class.

- If the equivalence classes are chosen wisely this greatly reduces the potential redundancy among test cases.

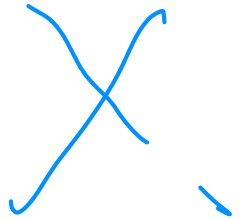
- The key point in equivalence class testing is the choice of the equivalence relation that determines the classes (partitions).

# Equivalence Class Selection



- tends to much of a “craft”:
  - no dependence on knowledge of code, only the specification
  - needs knowledge of input domain that usually goes beyond what an interface design specification provides
  - must understand how inputs are mutually dependent

# Equivalence Class Testing for 2-variable function



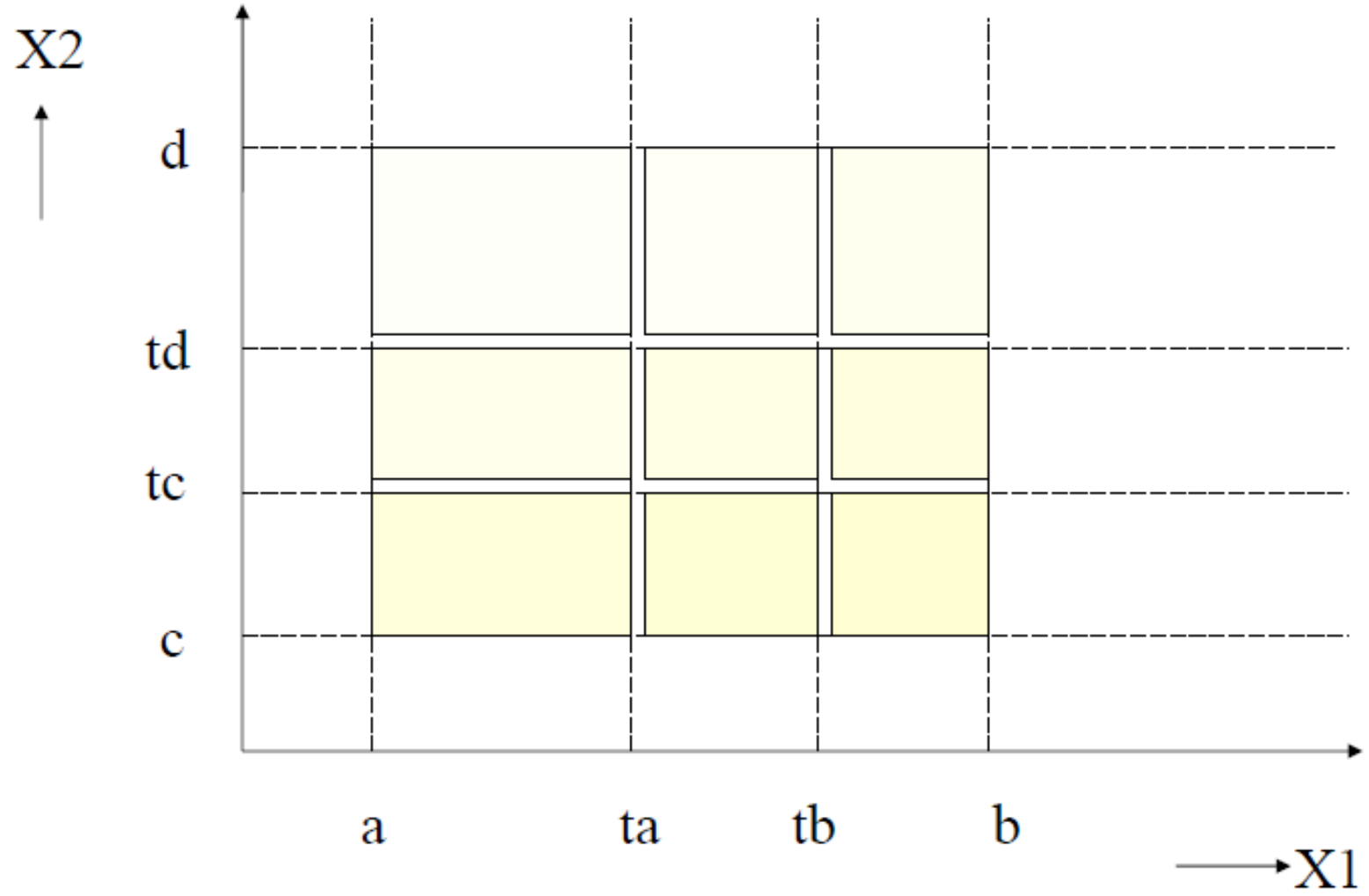
- Consider a function f(x1,x2) where the values of x1 and x2 are defined to be

$$a <= x1 <= b \quad and \quad c <= x2 <= d$$

- Assume the following equivalence classes for x1
  - *{a, a+1, …, ta}, {ta+1, ta+2, …, tb}, {tb+1, tb+2, …, b}*

- Assume the following equivalence classes for x2
  - *{c, c+1, …, tc}, {tc+1, tc+2, …, td}, {td+1, td+2, …, d}*

# Graphical Representation



# Test Data selection for 2-variable function

- Choose one value from each shaded region
  - Each value serves as a representative for all values in that region

- Total number of values = n * m where
  - ‘n’ is the number of partitions for input X1
  - ‘m’ is the number of partitions for input X2

# Equivalence Class Testing for 2-variable function – Robustness Testing

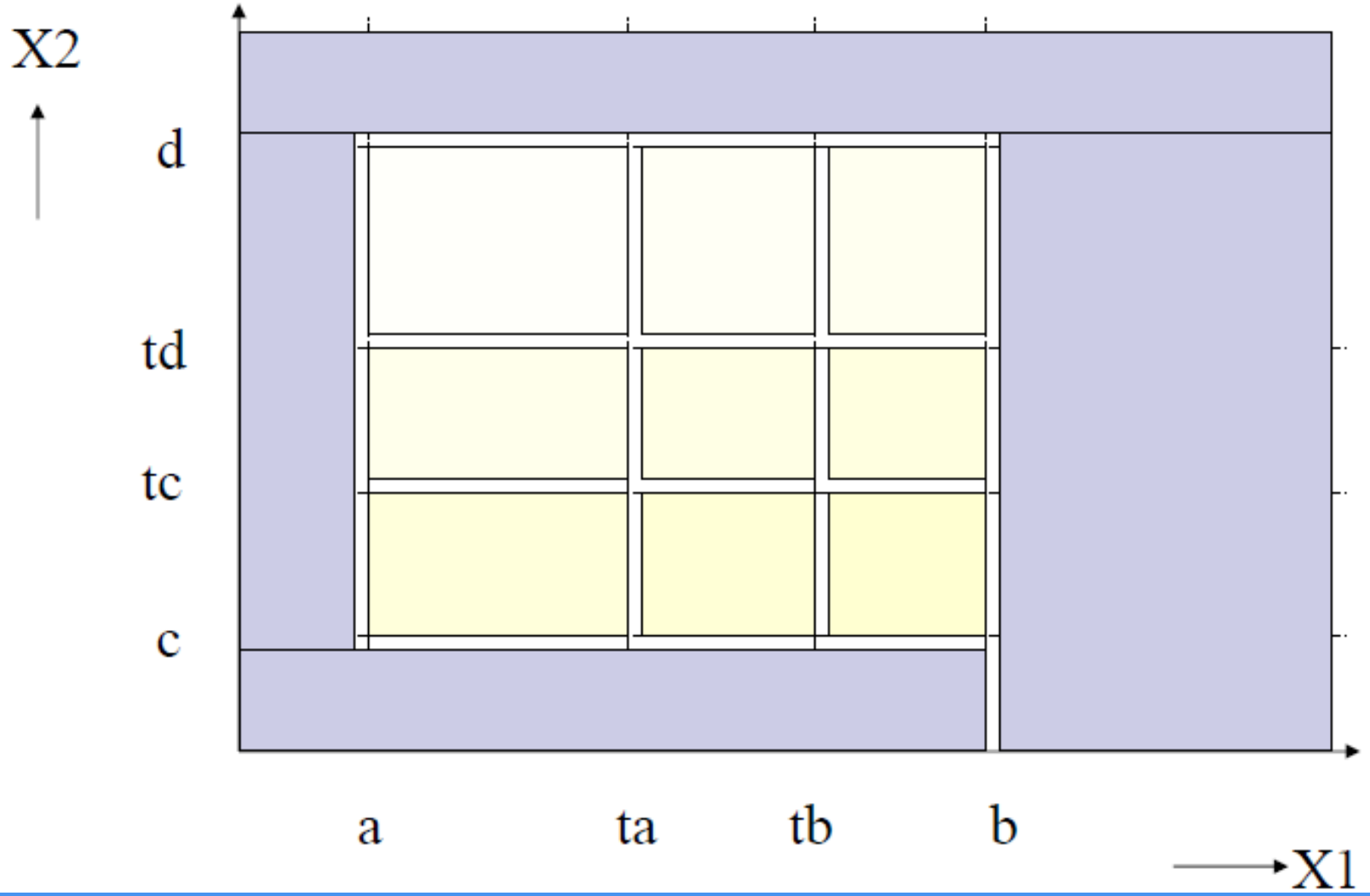
- Extended equivalence classes for x1

*{values < a}, {a, a+1, …, ta}, {ta+1, ta+2, …, tb}, {tb+1, tb+2, …, b}, {values > b}*

- Extended equivalence classes for x2

*{values < c}, {c, c+1, …, tc}, {tc+1, tc+2, …, td}, {td+1, td+2, …, d}, {values > d}*

# Graphical Representation



# Weak Normal Equivalence Class Testing

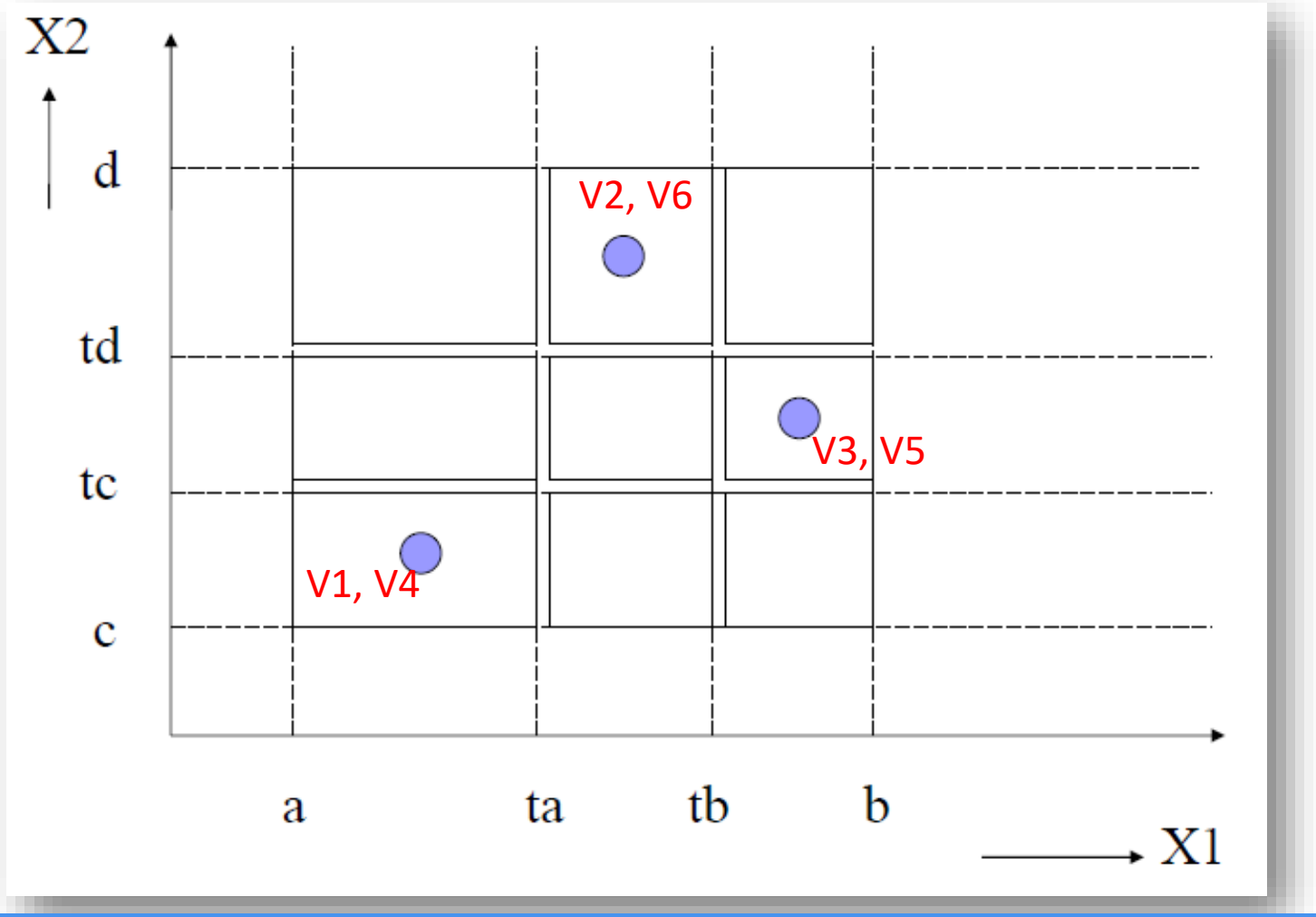
- In weak equivalence class testing, the aim is to cover all the equivalence classes at least once, and to do this we don't need to cover all possible combinations.

- The minimum number of test cases is equal to the number of classes in the partition with the largest number of subsets.

$n_x = 3 \quad n_y = 5 \Rightarrow 5$

# Weak Normal Equivalence Class Testing

- Variable x1
  - *V1 = {a, a+1, …, ta}*
  - *V2 = {ta+1, ta+2, …, tb}*
  - *V3 = {tb+1, tb+2, …, b}*
- Variable x2
  - *V4 = {c, c+1, …, tc}*
  - *V5 = {tc+1, tc+2, …, td}*
  - *V6 = {td+1, td+2, …, d}*



# Strong Normal Equivalence Class Testing

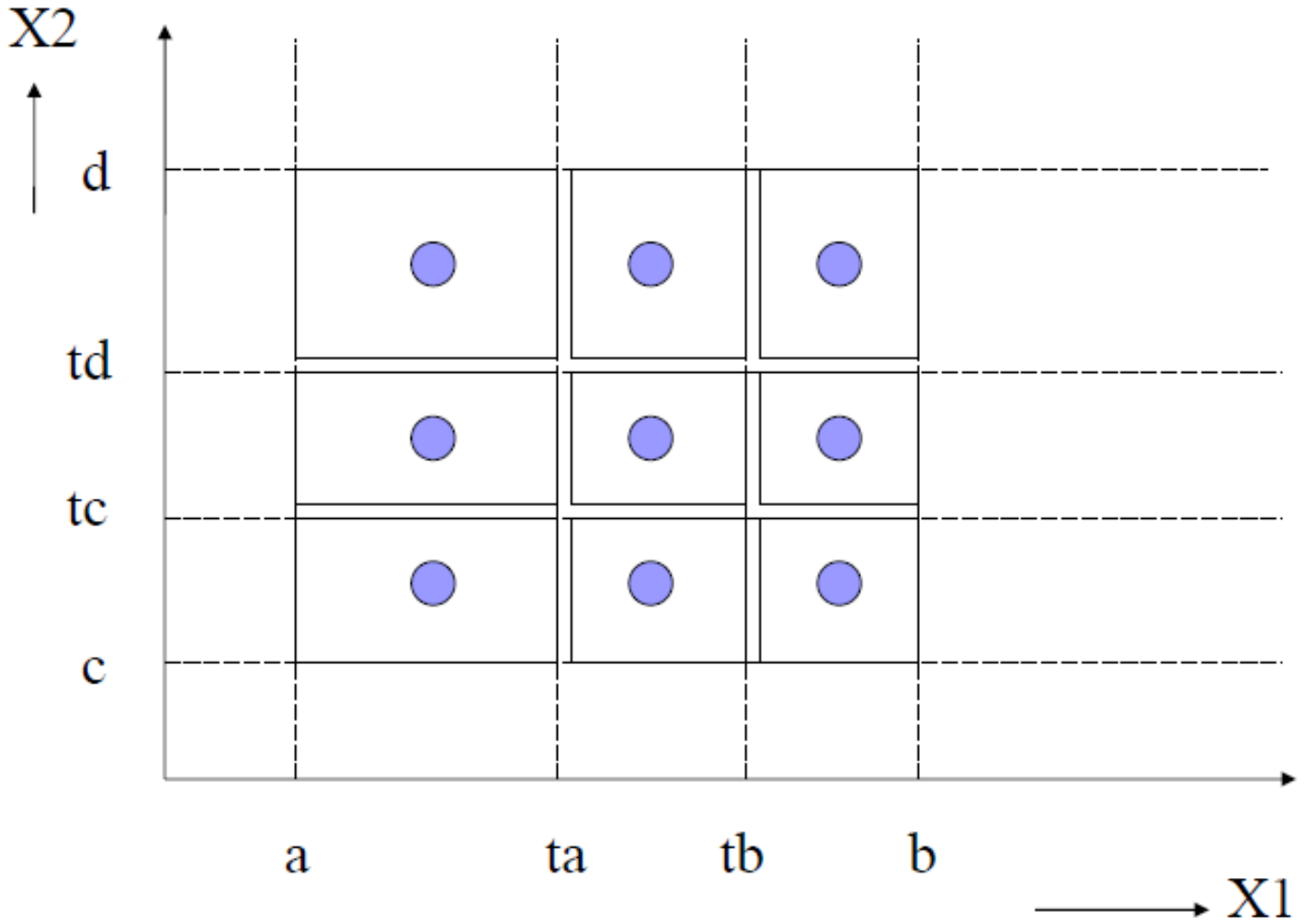
- Strong equivalence class testing is based on the Cartesian Product of the partition subsets.

- From the previous example, this would generate:
  - 3 * 3 = 9 test cases

- Generates more test cases which test for any interaction between the representative values from each of the subsets.

# Example

- Variable x1
  - *V1 = {a, a+1, …, ta}*
  - *V2 = {ta+1, ta+2, …, tb}*
  - *V3 = {tb+1, tb+2, …, b}*
- Variable x2
  - *V4 = {c, c+1, …, tc}*
  - *V5 = {tc+1, tc+2, …, td}*
  - *V6 = {td+1, td+2, …, d}*
- Test cases
  - <V1, V4>, <V1, V5>, <V1, V6>
  - <V2, V4>, <V2, V5>, <V2, V6>
  - <V3, V4>, <V3, V5>, <V3, V6>

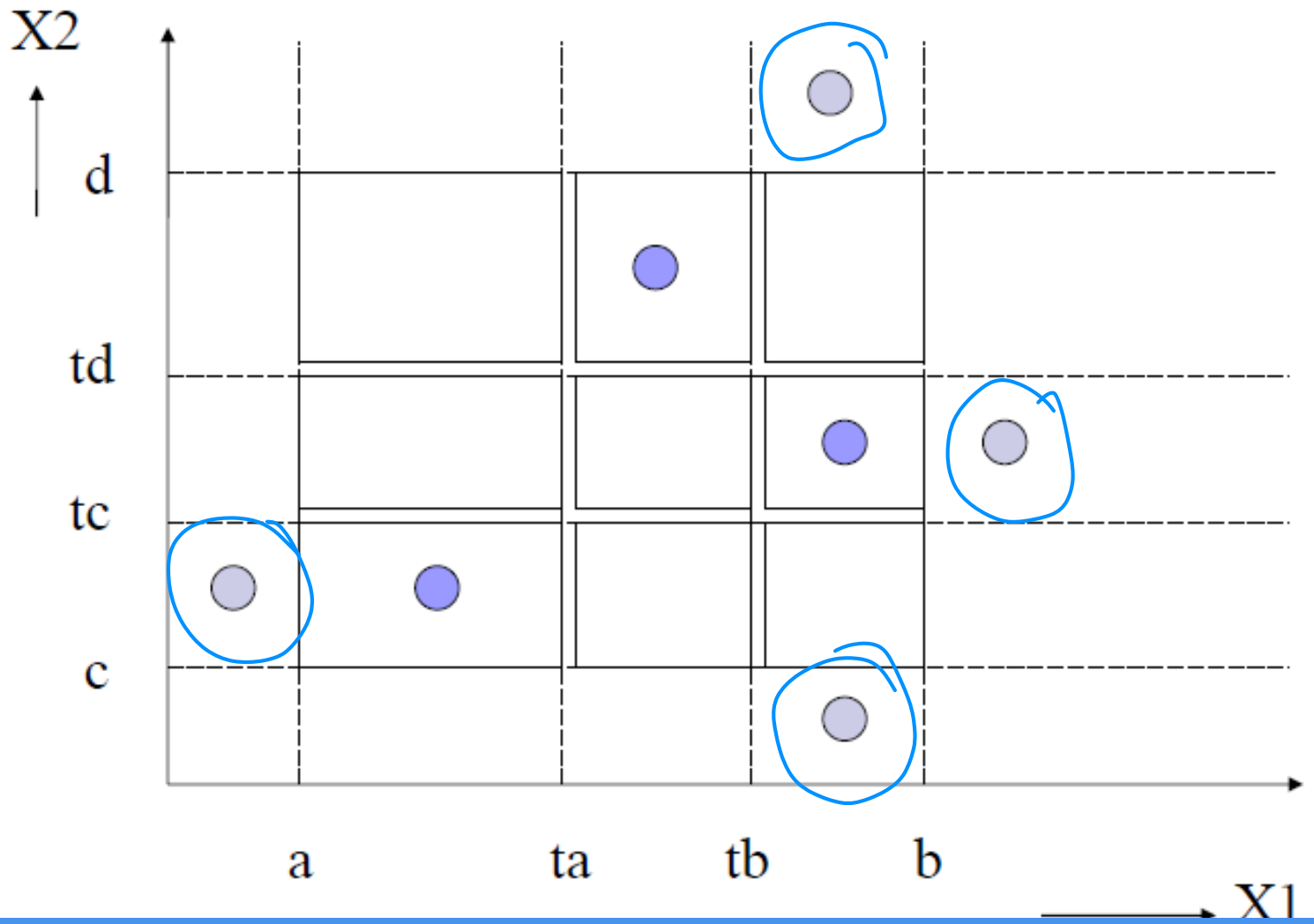


# Weak and Strong Robust Equivalence Class Testing

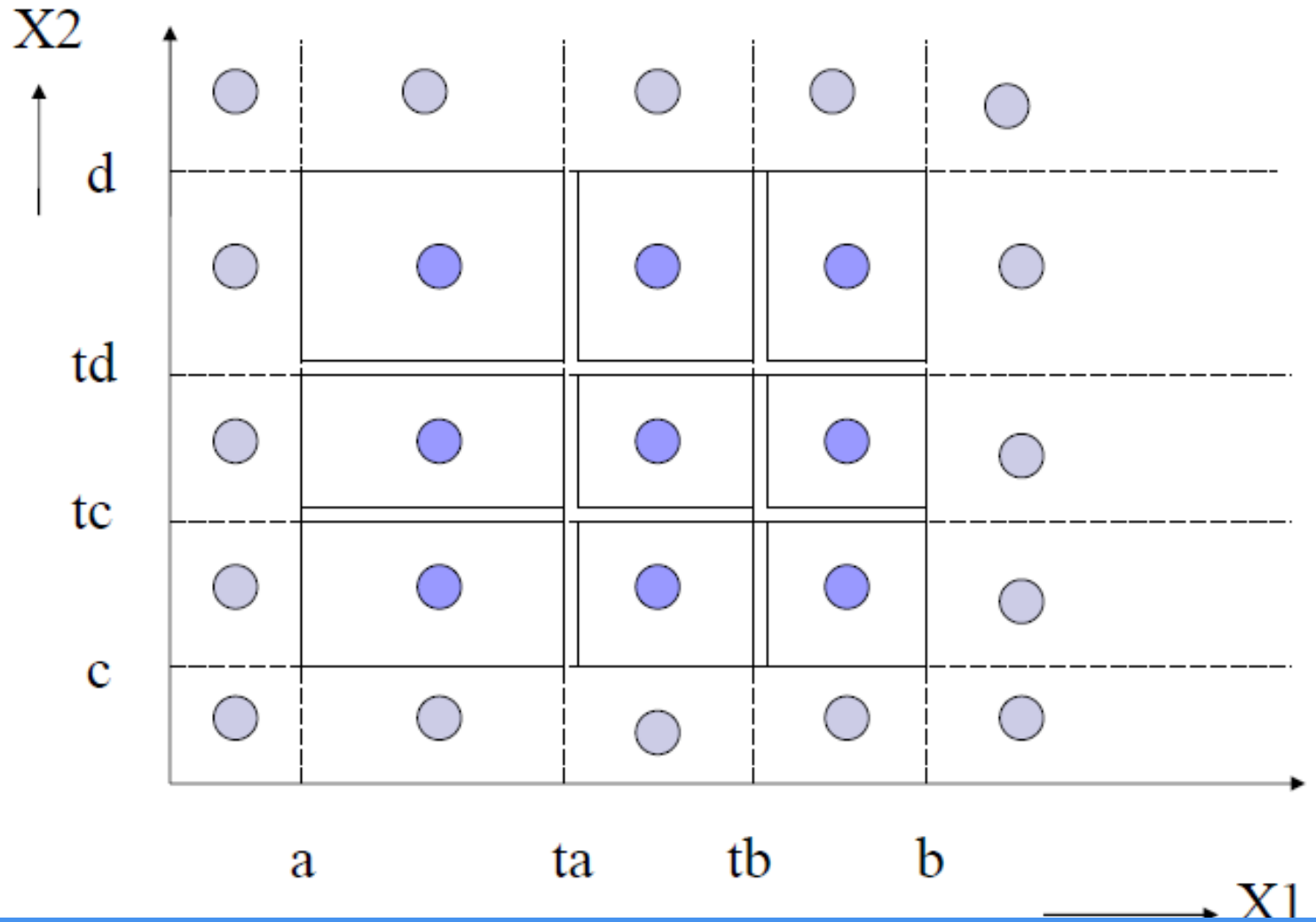
- Weak Robust Equivalence Class Testing
  - Similar to Weak Normal, except additional test cases are chosen from equivalence classes outside the range
  - One test case to cover out of range values for each variable

- Strong Robust Equivalence Class Testing
  - Similar to Strong Normal, except additional equivalence classes are added to cover out of range values
  - Test cases once again derived from the Cartesian product of all equivalence classes

# Weak Robust Equivalence class Testing



# Strong Robust Equivalence class Testing



# Example

- Consider a basic system that accepts two integer inputs, X and Y, and performs a division operation X/Y.
- Here are some equivalence classes for both X and Y:
  - Variable X: {values <= -1}, 0, {values >= 1}
  - Variable Y (denominator): {values <= -1}, 0 (invalid), {values >= 1}

# Example

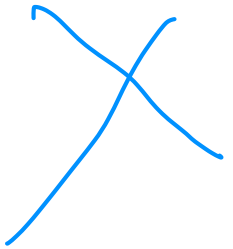
- Consider a basic system that accepts two integer inputs, X and Y, and performs a division operation X/Y.
- Here are some equivalence classes for both X and Y:
  - Variable X: {values <= -1}, 0, {values >= 1}
  - Variable Y (denominator): {values <= -1}, 0 (invalid), {values >= 1}

- Weak Normal Equivalence Class test cases
  - (X, Y) = (-3, -5), (0, 8), (12, 2)

# Output Equivalence Class

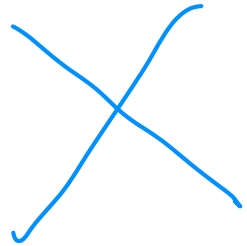
- Equivalence class partitioning can be applied also to the output domain of the software under test.
- Represent the different responses or behaviors that the software may exhibit based on its input.
- Verify whether the software behaves as expected for each class of outputs, which can be more efficient than testing every individual output.

# Output Equivalence Class Example



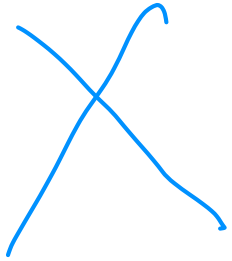
- Consider an example of a simple car rental system where the desired output is the rental cost.
- The system calculates the cost based on the type of car, rental duration (in days), and whether the customer has a membership for a discount.
- Here are the rules:
  1. Economy cars cost $20 per day, SUVs $50 per day, and Luxury cars are $100 per day.
  2. Members get a 10% discount on the total cost.

# Output Equivalence Class Example



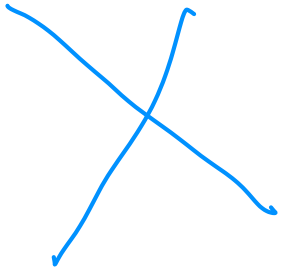
- The output equivalence classes here would be:
  1. Economy car cost for non-members: $20/day
  2. Economy car cost for members: $20/day with a 10% discount
  3. SUV cost for non-members: $50/day
  4. SUV cost for members: $50/day with a 10% discount
  5. Luxury car cost for non-members: $100/day
  6. Luxury car cost for members: $100/day with a 10% discount

# Output Equivalence Class Example



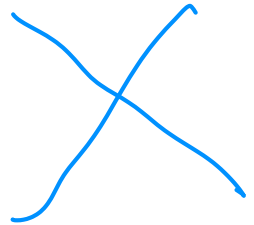
- Weak Normal Equivalence Class Testing, we would choose one representative from each equivalence class and test.
  1. Renting an Economy car for 3 days by a non-member.
  2. Renting an Economy car for 2 days by a member.
  3. Renting an SUV for 4 days by a non-member.
  4. Renting an SUV for 7 days by a member.
  5. Renting a Luxury car for 1 day by a non-member.
  6. Renting a Luxury car for 5 days by a member.

# Output Equivalence Class Example



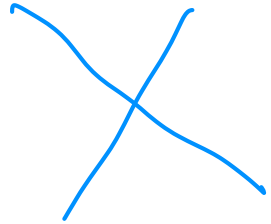
- Strong Normal Equivalence Class Testing, you would now choose one representative from each equivalence class and test for all possible combinations. Here are some representative test cases:
  1. Economy car rented by a non-member for 1 day.
  2. Economy car rented by a non-member for 2 days.
  3. Economy car rented by a member for 1 day.
  4. Economy car rented by a member for 2 days.
  5. SUV rented by a non-member for 1 day.
  6. SUV rented by a non-member for 2 days.
  7. SUV rented by a member for 1 day.
  8. SUV rented by a member for 2 days.
  9. Luxury car rented by a non-member for 1 day.
  10. Luxury car rented by a non-member for 2 days.
  11. Luxury car rented by a member for 1 day.
  12. Luxury car rented by a member for 2 days.

# Output Equivalence Class Example



- Robust Equivalence Class Testing involves testing with both valid and 'just outside' invalid inputs.
- In our car rental example, the valid inputs are:
  1. Type of car: Economy, SUV, or Luxury.
  2. Duration of rental: A positive integer for rental days.
  3. Membership status: Whether the customer is a member or not.
- And the invalid inputs could be:
  1. Type of car: Any type other than Economy, SUV, or Luxury (e.g., compact, pickup).
  2. Duration of rental: Non-positive integers (e.g., 0 or negative numbers).
  3. Membership status: Any status other than member or non-member (e.g., pending).

# Effective coverage of test cases

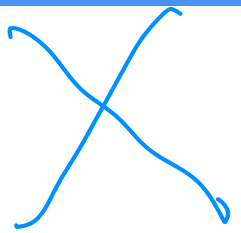


- It is preferable to apply equivalence class testing for both input domain and output domain
  - In practice, many people apply the technique only for input domain
- It is preferable to include “Robust equivalence classes”
- It is preferable to combine equivalence class testing with boundary value analysis and apply worst-case scenario (or multiple-fault assumption – more than one variable may be at fault)

# Scenario for Equivalence Class Testing

1. Large Input Domain: When the system under test has a large number of inputs, it's impractical to test them all. Equivalence class testing allows you to divide the input into groups (equivalence classes) and test a representative input from each group, greatly reducing the number of test cases.
2. Boundary Conditions: Equivalence class testing is useful for identifying boundary conditions where defects often occur. When you divide the inputs into valid and invalid classes, the boundaries between these classes often reveal errors.
3. Reducing Redundancy: If you believe that testing one value from an equivalence class is equivalent to testing all other values in that class, this method can eliminate redundant or duplicate testing and save considerable time and effort.

# Scenario for Equivalence Class Testing



1. Software with Defined Inputs: It works best when the software has clearly defined input domains that can be easily divided into classes. For example, choosing a country from a drop-down list or entering an age in a text box.
2. Quick and Early Bug Discovery: It enables faster testing and helps in identifying high-risk defects at an early stage of testing which improves the quality of the software product.

While equivalence class testing can identify many issues, it doesn't guarantee finding all errors, and should be used as part of a broader testing strategy that includes other techniques as well.

# Black-box Testing

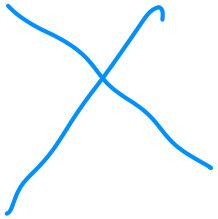
- Boundary Value Testing (BVT)
  - Normal boundary value testing
  - Robust boundary value testing
  - Worst-case boundary value testing
  - Robust worst-case boundary value testing
- Equivalence Class Testing
  - Weak Normal Equivalence Class Testing
  - Strong Normal Equivalence Class Testing
  - Weak Robust Equivalence Class Testing
  - Strong Robust Equivalence Class Testing
- **Decision Table Based Testing**

# Decision Table 决策表

- Decision tables are a precise yet compact way to model complicated logic.
- Decision tables, like if-then-else and switch-case statements, associate conditions with actions to perform.
- But, unlike the control structures found in traditional programming languages, decision tables can associate many independent conditions with several actions in an elegant way.

“http://en.wikipedia.org/wiki/Decision_table”

# Decision Table



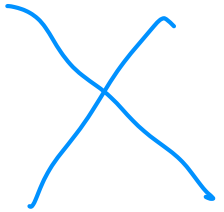
- Decision tables make it easy to observe that all possible conditions are accounted for.
- Decision tables can be used for Specifying complex program logic.

| | |
|---|---|
| Conditions/Condition Stubs | Condition Entries |
| Actions/Action Stubs | Action Entries |

# Decision Table Structure

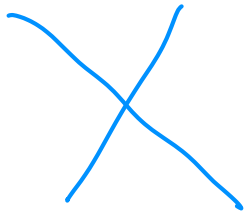
| *Stub* | *Rule 1* | *Rule 2* | *Rules 3, 4* | *Rule 5* | *Rule 6* | *Rules 7, 8* |
|---|---|---|---|---|---|---|
| c1 | T | T | T | F | F | F |
| c2 | T | T | F | T | T | F |
| c3 | T | F | — | T | F | — |
| a1 | X | X | | X | | |
| a2 | X | | | | X | |
| a3 | | X | | X | | |
| a4 | | | X | | | X |

# Decision Table Structure



- Each condition corresponds to a variable, relation or predicate
- Possible values for conditions are listed among the condition entries
  - Boolean values (True / False) – Limited Entry Decision Tables
  - Several values – Extended Entry Decision Tables
  - Don’t care value
- Each action is a procedure or operation to perform
- The Action entries specify whether the action is to be performed

# Decision Table Example



- Consider a function f(x1,x2) where the values of x1 and x2 are defined to be

$$a <= x1 <= b \quad and \quad c <= x2 <= d$$

- When both input values x1 and x2 are within their ranges (defined in the condition stubs), compute y value (described in the action stubs).
- In all other situations, output an error message

# Decision Table Example

| | | | | |
|---|---|---|---|---|
| a <= x1<= b | T | T | F | F |
| c <= x2 <= d | T | F | T | F |
| Compute y | X | | | |
| Output error | | X | X | X |

# Decision Table Example

| | | | | |
|---|---|---|---|---|
| a <= x1<= b | T | -- | F | F |
| c <= x2 <= d | T | F | -- | F |
| Compute y | X | | | |
| Output error | | X | X | X |

‘--’ indicates “don’t care”

# Decision Table Example

| | | | | | |
|---|---|---|---|---|---|
| x1< a | T | F | F | F | F |
| x1 = a | F | T | F | F | F |
| a < x1 < b | F | F | T | F | F |
| x1 = b | F | F | F | T | F |
| x1 > b | F | F | F | F | T |
| c <= x2 <= d | -- | T | T | T | -- |
| Compute y | | X | X | X | |
| Output error | X | | | | X |

# Decision Table Example – Extended Entry Decision Table

| | Conditions/ Courses of Action | Rules | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 |
| **Condition Stubs** | Employee type | S | H | S | H | S | H |
| | Hours worked | <40 | <40 | 40 | 40 | >40 | >40 |
| | | | | | | | |
| **Action Stubs** | Pay base salary | X | | X | | X | |
| | Calculate hourly wage | | X | | X | | X |
| | Calculate overtime | | | | | | X |
| | Produce Absence Report | | X | | | | |

# Decision Table Development Methodology

1. Determine conditions and values
2. Determine maximum number of rules
3. Determine actions
4. Encode possible rules
5. Encode the appropriate actions for each rule
6. Verify the policy
7. Simplify the rules (reduce the number of columns whenever possible)

# Decision Table - Usage

- The use of the decision-table model is applicable when :
  - The specification is given or can be converted to a decision table.
  - The order in which the predicates are evaluated does not affect the interpretation of the rules or resulting action.
  - The order of rule evaluation has no effect on resulting action.
  - Once a rule is satisfied and the action selected, no other rule need be examined.
  - The order of executing actions in a satisfied rule is of no consequence.

# Decision Table -Issue

Before using the tables, ensure:

- rules must be complete
  - every combination of predicate truth values plus default cases are explicit in the decision table
- rules must be consistent
  - every combination of predicate truth values results in only one action or set of actions

# Inconsistent Decision Table 不一致的决策表.

| **Conditions** | 1 -4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|
| C1 | T | F | F | F | F | T |
| C2 | - | T | T | F | F | F |
| C3 | - | T | F | T | F | F |
| **Actions** | | | | | | |
| a1 | X | X | X | - | - | - |
| a2 | - | X | X | X | - | X |
| a3 | X | - | X | X | X | - |

# Decision Table for Triangle Problem

判断三角形

| **Conditions** | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| C1: a < b+c? | F | T | T | T | T | T | T | T | T | T | T |
| C2: b < a+c? | - | F | T | T | T | T | T | T | T | T | T |
| C3: c < a+b? | - | - | F | T | T | T | T | T | T | T | T |
| C4: a=b? | - | - | - | T | T | T | T | F | F | F | F |
| C5: a=c? | - | - | - | T | T | F | F | T | T | F | F |
| C6: b=c? | - | - | - | T | F | T | F | T | F | T | F |
| **Actions** | | | | | | | | | | | |
| A1: Not a Triangle | X | X | X | | | | | | | | |
| A2: Scalene | | | | | | | | | | | X |
| A3: Isosceles | | | | | | | X | | X | X | |
| A4: Equilateral | | | | X | | | | | | | |
| A5: Impossible | | | | | X | X | | X | | | |

# Test Cases for Triangle Problem

| Case ID | a | b | c | Expected Output |
|---|---|---|---|---|
| DT1 | 4 | 1 | 2 | Not a Triangle |
| DT2 | 1 | 4 | 2 | Not a Triangle |
| DT3 | 1 | 2 | 4 | Not a Triangle |
| DT4 | 5 | 5 | 5 | Equilateral |
| DT5 | ? | ? | ? | Impossible |
| DT6 | ? | ? | ? | Impossible |
| DT7 | 2 | 2 | 3 | Isosceles |
| DT8 | ? | ? | ? | Impossible |
| DT9 | 2 | 3 | 2 | Isosceles |
| DT10 | 3 | 2 | 2 | Isosceles |
| DT11 | 3 | 4 | 5 | Scalene |

# Observation

- Decision Table testing is most appropriate for programs where
  - Prominent if-then-else logic
  - there are important logical relationships among input variables
  - There are calculations involving subsets of input variables
  - There are cause and effect relationships between input and output
  - There is complex computation logic (high cyclomatic complexity)
- Decision tables do not scale up very well
- Decision tables can be iteratively refined

# End